

Luis Najera

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EDUCATION

National Taiwan University, Graduate School of Advanced Technology Taipei, Taiwan
Master of Science in Nanotechnology and Nanoscience Sep. 2024 – Expected Aug. 2026

- Thesis: A Multi-Modal Off-Axis Polarized Digital Holography Microscopy System for Depth-Resolved Stress Measurements in Advanced Packaging.

Universidad del Valle de Guatemala Guatemala City, Guatemala
Bachelor of Science in Electronics Engineering Jan. 2015 – Nov. 2019

- Thesis: HDL-to-Netlist Circuit Implementation for Integrated Circuit Design Flow.

EXPERIENCE

Graduate Research Assistant Sep. 2024 – Present
Precision Metrology Laboratory, National Taiwan University Taipei, Taiwan

- Developed a hybrid optical metrology platform for stress measurement and 3D profile reconstruction of Through-Glass Via (TGV) structures and transparent industrial samples.
- Designed CAD models, selected optical and optomechanical components, and implemented experimental setups integrating polarization imaging, digital photoelasticity, and off-axis/on-axis digital holography.
- Developed MATLAB-based image processing routines for holographic phase reconstruction, polarization analysis, retardation-to-stress mapping, and experimental data visualization.
- Performed optical measurements on real samples provided by Taiwan-based industry partners, supporting system validation for semiconductor, display, and transparent material inspection applications.
- Prepared and delivered technical presentations to external partners, summarizing system design, measurement capabilities, experimental results, and potential industrial applications.

R&D Intern – Optical Metrology Jul. 2025 – Aug. 2025
TRI Innovations Taipei, Taiwan

- Designed, assembled, aligned, and tested a reflection-mode off-axis digital holography system for optical metrology and 3D reconstruction applications.
- Selected optical and optomechanical components, configured the imaging system, and evaluated experimental performance under different acquisition conditions.
- Developed Python-based image processing routines for phase and amplitude reconstruction of measured samples.
- Prepared technical reports and presentations summarizing experimental progress, system limitations, and future improvement strategies.

R&D Engineer / Automation Engineer Sep. 2020 – Aug. 2023
Ingenio La Unión Escuintla, Guatemala

- Designed, developed, tested, and implemented electronic circuits and industrial automation systems for sugar manufacturing applications.
- Designed and reviewed PCB layouts, selected electronic components, and validated circuit performance for automation and control systems.
- Programmed and installed electronic systems used to improve industrial process monitoring, automation reliability, and operational efficiency.
- Collaborated with multidisciplinary teams to troubleshoot hardware, validate system performance, and support implementation in industrial environments.

PROJECTS

A Multi-Modal Off-Axis Polarized DHM System | *MATLAB, Digital Holography, Photoelasticity*

- Developed a multi-modal optical metrology system combining off-axis digital holography and digital photoelasticity for depth-resolved stress measurement and 3D profile reconstruction in Through-Glass Via (TGV) structures.
- Implemented phase reconstruction, polarization analysis, retardation-to-stress mapping, and 3D profile extraction routines for transparent advanced packaging samples.
- Integrated optical system design, experimental alignment, image acquisition, and computational post-processing into a unified metrology workflow.

Dark-Field Inspection System for Transparent Glass Components | *Optical Design, Bi-Telecentric Lens, CAD*

- Designed a dark-field optical inspection system using a bi-telecentric lens for defect visualization in transparent glass samples provided by an external consumer electronics partner.
- Selected optical and optomechanical components, prepared CAD models, and evaluated system configuration for large-field inspection of circular transparent samples.

Reflection-Mode Off-Axis Digital Holography System | *Python, Optical Metrology, 3D Reconstruction*

- Designed, assembled, aligned, and tested a reflection-mode off-axis digital holography system for optical metrology and 3D surface reconstruction.
- Developed Python-based routines for holographic phase and amplitude reconstruction, phase compensation, and experimental result visualization.

Embedded Electronics and Instrumentation Systems | *PCB Design, Embedded Systems, Industrial Automation*

- Designed PCB-based electronic systems for biomedical training and industrial monitoring applications, including sensor integration, circuit validation, and system-level testing.
- Hardware development for low-cost medical equipment and remote monitoring systems for industrial irrigation pumps.

PUBLICATIONS AND CONFERENCE PRESENTATIONS

L. Nájera et al., “Single-exposure Polarization Digital Holographic Microscopy (SPDHM) for Quantitative Stress Metrology in Laser-modified Glass Substrates for Advanced Packaging,” *ISOT 2026*. First author.

Surajit Das et al., “Quantitative Analysis of Numerical Refocusing in Digital Holographic Microscopy with Application to Through-Glass Via Metrology for Advanced Semiconductor Packaging,” *ISOT 2026*. Co-author.

Liang-Chia Chen et al., “Single-Shot Polarization-Phase-Shifting Digital Holography Microscopy for Through-Glass Via Critical Dimensions and Surface Defects Metrology,” *PRESM 2026*. Poster presentation, co-author.

Bo-Chen Kuo et al., “SWIR Oblique-Scanning Scatterometry with Global Sensitivity Analysis and Back Focal Plane Imaging for Hidden HAR Microstructure Critical Dimension Measurement,” *Proceedings of SPIE*, 135671M. Co-author.

HONORS AND AWARDS

Taiwan MOFA Scholarship, awarded by the Ministry of Foreign Affairs of Taiwan to pursue graduate studies in Taiwan. 2023 – 2026

Graduation Project Excellence Award, Universidad del Valle de Guatemala.

Academic Merit Honor for outstanding academic achievement, Universidad del Valle de Guatemala.

First Place, 2019 Scientific Fair, Universidad del Valle de Guatemala.

Second Place, 2018 Scientific Fair, Universidad del Valle de Guatemala.

PATENTS

Hybrid Polarized Digital Holographic Microscopy System for Surface Topography and Stress Characterization. Patent application filed, National Taiwan University.

TECHNICAL SKILLS

Optical Metrology: Digital holographic microscopy, polarization imaging, digital photoelasticity, reflectometry, phase reconstruction, stress mapping, 3D profile reconstruction, dark-field imaging

Programming and Data Analysis: Python, MATLAB, image processing, numerical analysis, experimental data visualization

Simulation and Modeling: Ansys Lumerical FDTD/RCWA, reflectometry-based modeling, simulation-to-experiment matching, parameter extraction

Electronics and Automation: Electronic circuit design, PCB layout, embedded systems, industrial automation

LANGUAGES

Spanish: Native

English: C2 / Full professional proficiency

French: B2 / Upper-intermediate

Mandarin Chinese: A2 / Elementary proficiency